

IT Security 2

Unit 6 – Security Policy Rules

Instructor: Brian Krause

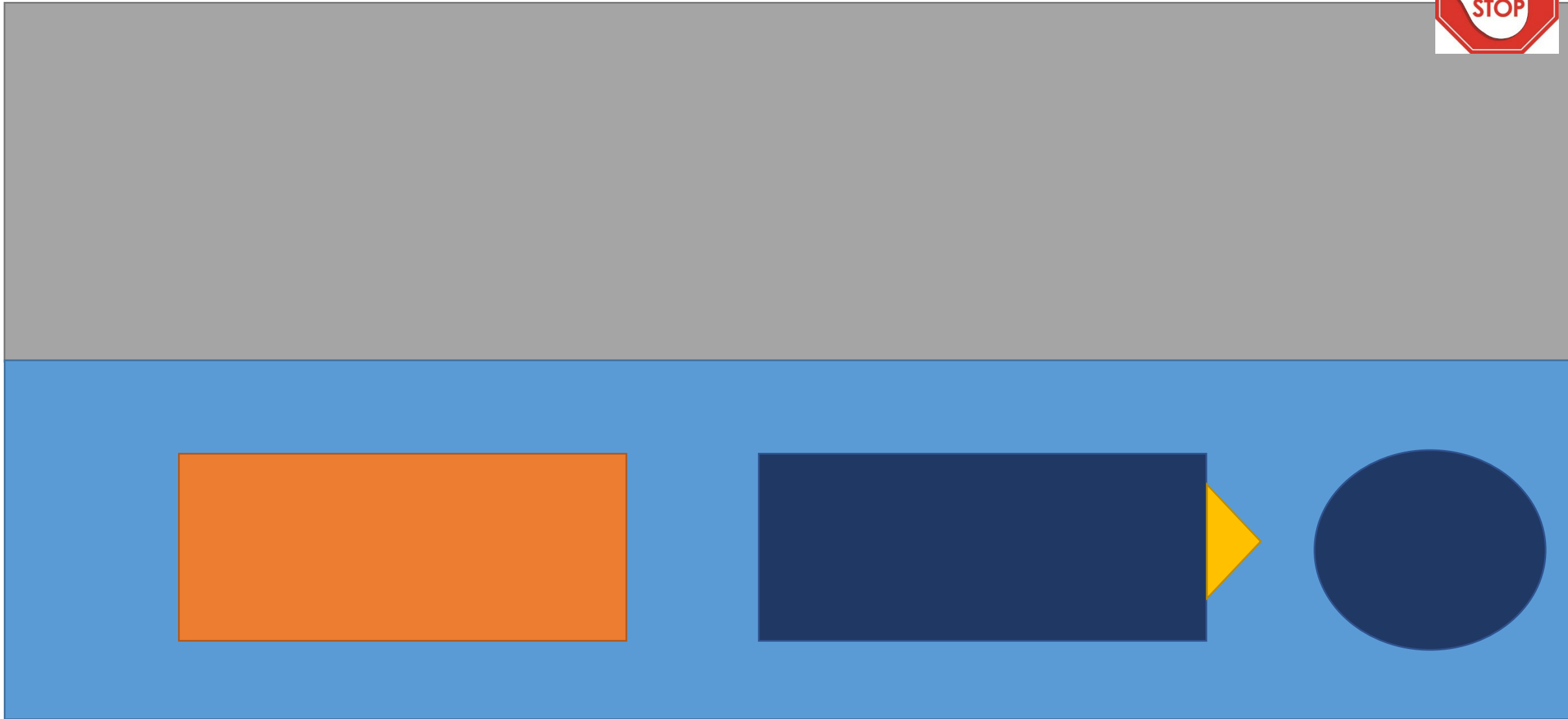
Announcements

- Next Week Class - NAT

Security Policy Rules

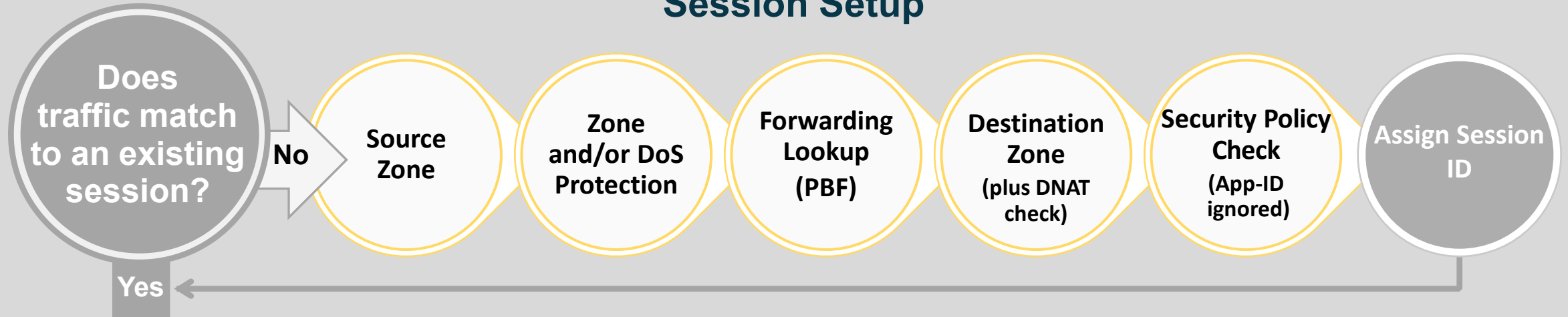
Some slides were adapted from Palo-Alto for use in this class. Content is a mesh between vendor provided information and instructor comments.

Flow Logic of the Next-Generation Firewall

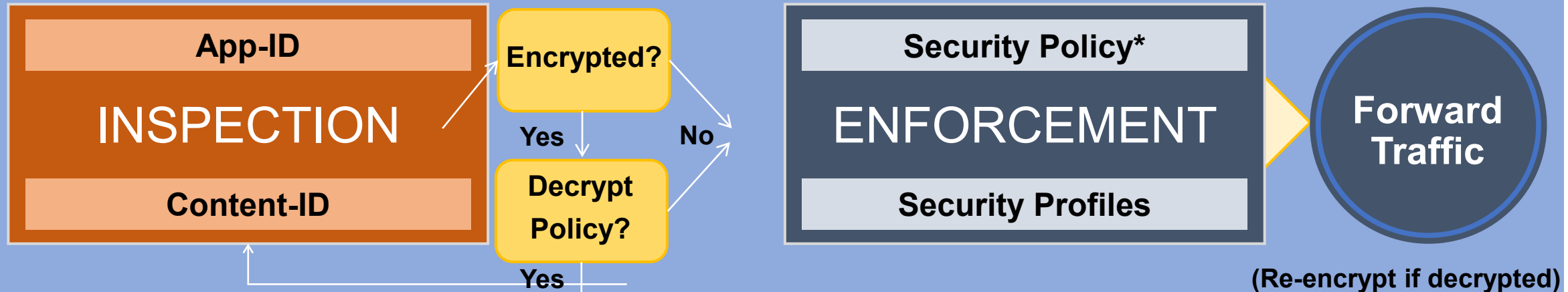


Flow Logic of the Next-Generation Firewall

Session Setup

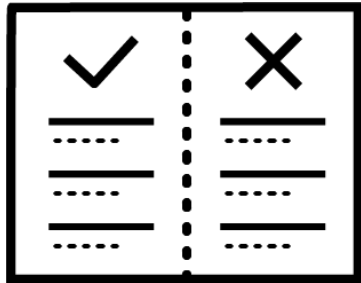


Inspection and Enforcement



* Policy check relies on pre-NAT IP addresses

CREATING AND MANAGING SECURITY POLICY RULES



GET TRAFFIC FLOWING

- Security policy fundamental concepts
- Security policy administration

Learning Objectives

After you complete this module, you should be able to:

- Describe Security policy concepts and operation
- Configure a Security policy rule
- Manage a Security policy
- Create and use tags and custom services in a Security policy





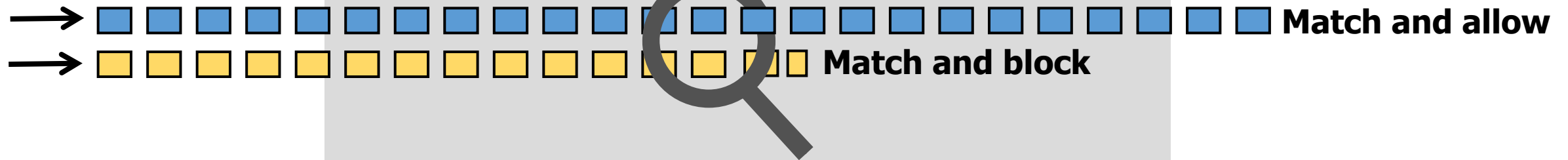
Security policy fundamental concepts

Security policy administration

Inspect and Control Network Traffic

Multiple match criteria available to control network traffic

Network packets



Basic Criteria

- Source zone
- Destination zone

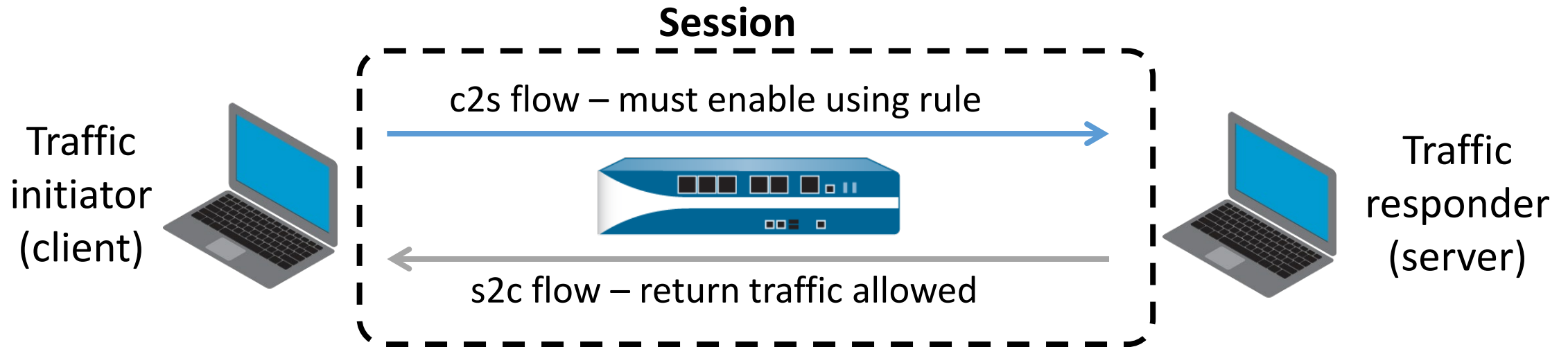
More Granular Criteria

- Source address
- Destination address
- Port
- Application
- URL category
- Source user
- HIP Profile

Fields in Security policy rules

Firewall Data Plane

Sessions and Flows



- A packet is matched to a session; each session is matched to a Security policy rule.
- A session can consist of one or two flows:
 - Single flow example: multicast traffic
 - Two flow example: TCP traffic
- Server definition for a firewall is different from server definition for hosts:
 - Traffic responder versus providing a service

Display Security Policy Rules

Policies > Security

	NAME	TAGS	TYPE	Source				Destination			APPLICATION	SERVICE	ACTION
				ZONE	ADDRESS	USER	DEVICE	ZONE	ADDRESS	DEVICE			
1	Users_to_Extranet	Users_Net	universal	Users_Net	any	any	any	Extranet	any	any		application-defa...	Allow
2	Users_to_Internet	Users_Net	universal	Users_Net	any	any	any	Internet	any	any			
3	Extranet_to_Internet	Extranet	universal	Extranet	any	any	any	Internet	any	any			
4	intrazone-default	none	intrazone	any	any	any	any	(intrazone)	any	any	any		
5	interzone-default	none	interzone	any	any	any	any	any	any	any	any		

Columns
Adjust Columns

- ☒ Name
- ☒ Tags
- ☐ Group
- ☒ Type
- ☒ Source Zone
- ☒ Source Address
- ☒ Source User
- ☒ Source Device
- ☒ Destination Zone
- ☒ Destination Address
- ☒ Destination Device
- ☒ Application
- ☒ Service
- ☒ Action
- ☐ URL Category
- ☒ Profile
- ☒ Options

- Display and manage Security policy rules using the web interface.
- Click any column header to change the number of displayed columns:
 - Customized per user
- The list order matches the column order displayed in the web interface.

Manage the Policy Ruleset

Policies > Security

	NAME	TAGS		DEVICE					APPLICATION	SERVICE	ACTION		
1	Users_to_Extranet	Users_Net	universal	Users_Net	any	any	any	Extranet	any	any	any	application-defa...	Allow
2	Users_to_Internet	Users_Net	universal	Users_Net	any	any	any	Internet	any	any	any	application-default	Allow
3	Extranet_to_Internet		universal	Extranet	any	any	any	Internet	any	any	any	application-defa...	Allow
4	intrazone-default		intrazone				(intrazone)	any	any	any	any	any	Allow
5	interzone-default		interzone				any	any	any	any	any	any	Deny

Filter

Log Viewer

Move

Copy UUID

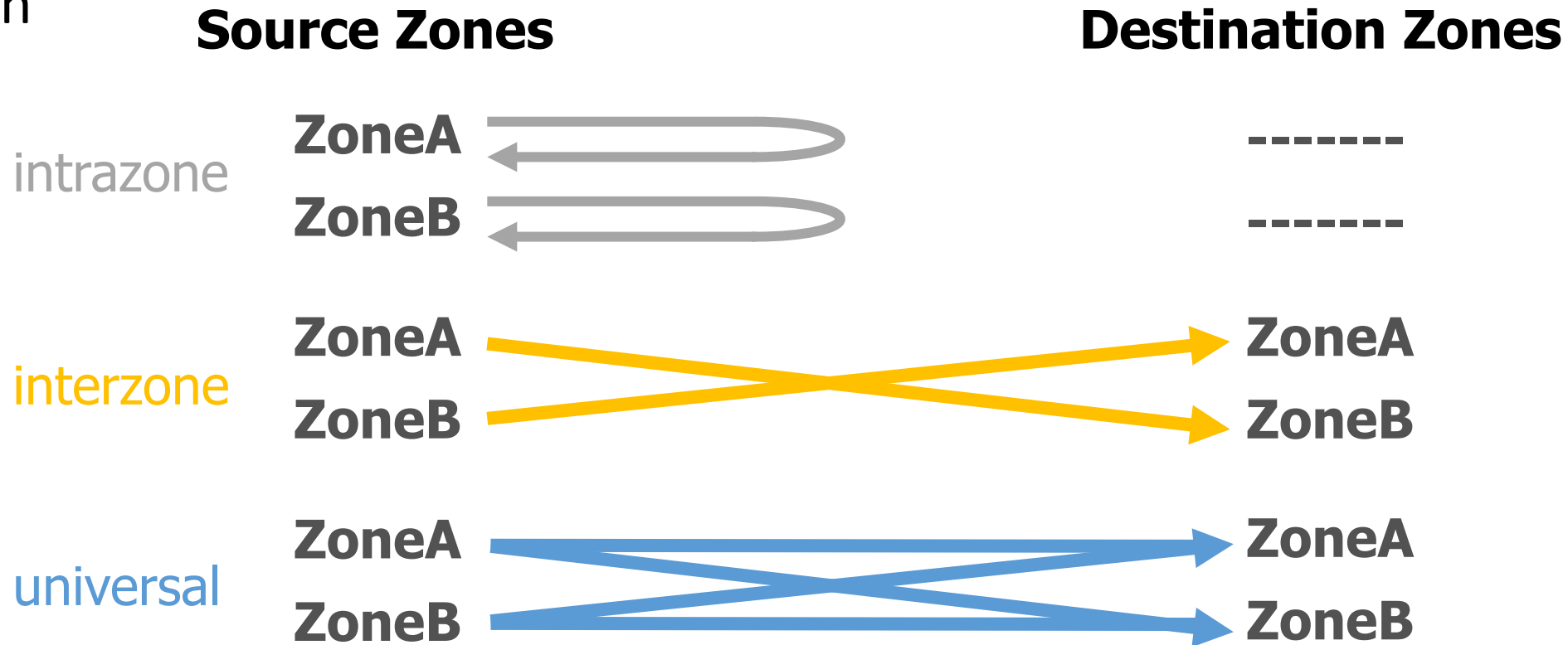
Global Find

+ Add - Delete Clone Override Revert Enable Disable Move PDF/CSV Highlight Unused Rules View Rulebase as Groups Reset Rule Hit Counter Group

- **Add, Delete, Clone, Override, Revert, Enable, Disable, Move** options.
- Rules can be re-ordered to match requirements (use **Move** or drag-and-drop).
- Disabling of a rule allows you to retain the entry while making it non-operative.

Security Policy Rule Types

- Three rule types
- Specifies whether a rule applies to traffic within a zone, between zones, or both



Custom and Predefined Rules

- By default, the firewall implicitly allows intrazone traffic and denies interzone traffic.
- Create explicit rules to control all other traffic.

Custom rule: By default, traffic is logged.

Predefined rules: By default, traffic is not logged.

	NAME	TAGS	TYPE	ZONE	ADDRESS				DEVICE	APPLICATION	SERVICE	ACTION
1	Users_to_Extranet	Users_Net	universal	Users_Net	any				any	any	application-defa...	Allow
2	Users_to_Internet	Users_Net	universal	Users_Net	any	any	any	Internet	any	any	application-default	Allow
3	Extranet_to_Internet	Extranet	universal	Extranet	any	any	any	Internet	any	any	application-defa...	Allow
4	intrazone-default	none	intrazone	any	any	any	any	(intrazone)	any	any	any	Allow
5	interzone-default	none	interzone	any	any	any	any	any	any	any	any	Deny

+ Add - Delete Clone Override Revert Enable Disable Move

PDF/CSV Highlight Unused Rules View Rulebase as Groups Reset Rule Hit Counter Group

Security Policy Rule Match

- Rules evaluated from top to bottom
- Further rules not evaluated after a rule match

	NAME	TAGS	TYPE	Source		Destination		APPLICATION	SERVICE	ACTION
				ZONE	ADDRESS	ZONE	ADDRESS			
1	Rule A	Egress	universal	Inside	any	Outside	any	web-browsing	any	Allow
2	Rule B	Egress	universal	Guest	any	Outside	any	web-browsing	any	Allow
3	Rule C	Egress	universal	DMZ	any	Outside	any	ftp	application-default	Allow
4	Rule D	Egress	universal	Inside	192.168.1.3	Outside	any	any	any	Allow

- Could Rule A and Rule B be combined?
- Yes:
 - Place Inside and Guest together in source zone.
 - Outside remains in destination zone.

Policy Rule Hit Count

- Identify rules that are frequently or seldom used.
- Determine the first time and last time a rule was used.
- View number of applications seen by a rule.
- Can be used to verify configuration changes.

	NAME	TAGS	TYPE	Source		Destination		APPLICATION	SERVICE	ACTION	Rule Usage			
				ZONE	ADDRESS	ZONE	ADDRESS				HIT COUNT	LAST HIT	FIRST HIT	APPS SEEN
1	Users_to_Internet	Internet	universal	Users_Net	any	Internet	any	dns google-base shutterfly ssl web-browsing	application-default	Allow	386	2020-07-14 18:12:31	2020-07-14 17:58:58	-
2	Users-Extranet	Extranet	universal	Users_Net	any	Extranet	any	any	application-default	Allow	401	2020-07-14 18:12:21	2020-07-14 17:59:43	1
3	Extranet_to_Internet	none	universal	Extranet	any	Internet	any	any	application-default					-
4	intrazone-default	none	intrazone	any	any	(intrazone)	any	any	any				7:10:46	-
5	interzone-default	none	interzone	any	any	any	any	any	any					-

Timestamp of first policy rule match and last policy rule match

Reset
Last Reset Time >

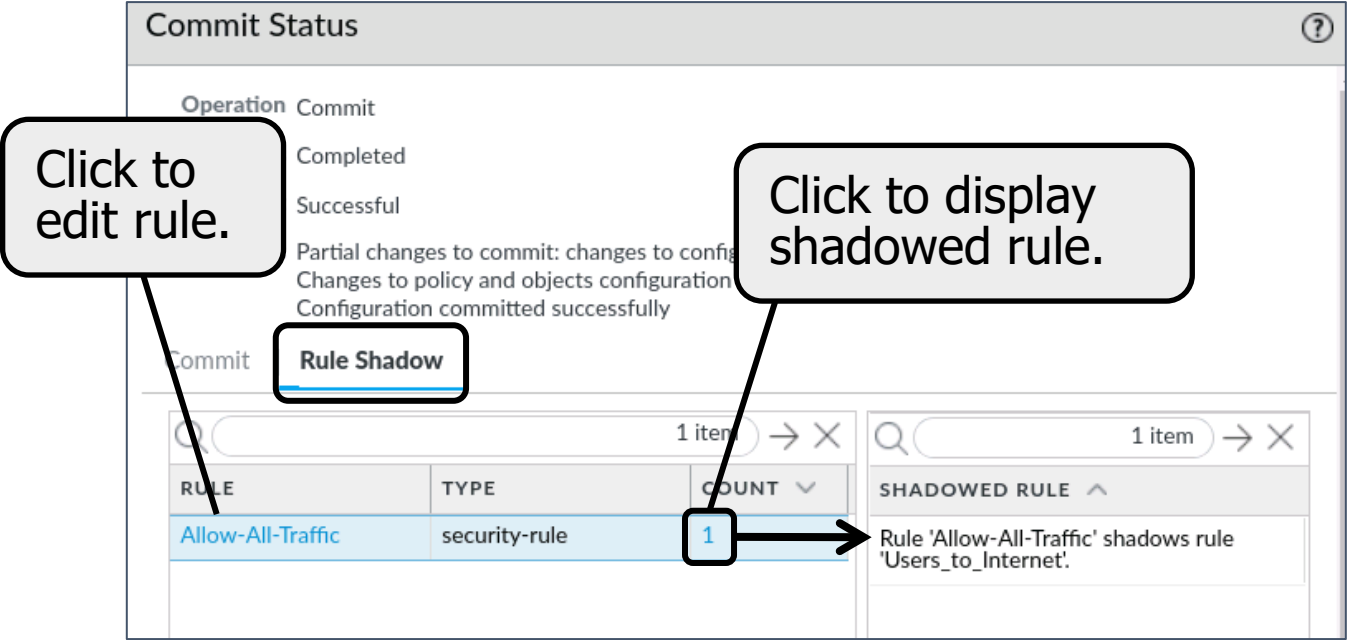
Number of applications seen by this rule

All rules
Selected rules

+ Add - Delete Clone Override Revert Enable Disable Move PDF/CSV Highlight Unused Rules View Rulebase as Groups Reset Rule Hit Counter Group Test Policy Match

Rule Shadowing

- Traffic can match multiple rules.
- Earlier rule hides (casts a shadow) over later rule.
- **Rule Shadow** tab in **Commit Status** window reports shadowed rules.
- Reorder or refine rules to remove shadowing.



	NAME	TAGS	TYPE	Source				Destination			APPLICATION	SERVICE	ACTION
				ZONE	ADDRESS	USER	DEVICE	ZONE	ADDRESS	DEVICE			
1	Allow-All-Traffic	Internet	universal	Users_Net	any	any	any	Internet	any	any	any	application-default	Allow
2	Users_to_Internet	Internet	universal	Users_Net	any	any	any	Internet	any	any	dns google-base shutterfly ssl web-browsing	application-default	Allow

Security policy fundamental concepts



Security policy administration

Configure a Security Policy Rule: General Tab

Policies > Security > Add

Security Policy Rule

General | Source | Destination | Application | Service/URL Category | Actions | Usage

Name: Users_to_Internet

Rule Type: universal (default)

Description: Allows hosts in Users_Net zone to access Internet

Tags: Internet

Group Rules By Tag: Internet

Audit Comment: Policy initially created by admin-bob on 7/24

[Audit Comment Archive](#)

universal (default)
intrazone
interzone

Usage tab appears after policy rule is created.

Optional, for easier visual identification and web interface filtering

Add audit comment describing what was added, when, and by whom.

View the audit comments, configuration logs, and rule change history.

Rule Changes Archive

Security Policy Rule

General | Source | Destination | Application | Service/URL Category | Actions

Name: Users_to_Internet
Rule Type: universal (default)
Description: Allows hosts in Users_Net

Tags: Internet X
Group Rules By Tag: Internet

Audit Comment: Added A/V, A/S, and URL filtering

[Audit Comment Archive](#)

Audit Comment Archive for Security Rule Users_to_Internet

Audit Comments | Config Logs (between commits) | **Rule Changes**

219 Committed On 2020/07/14 20:41:12 by admin

Select a rule config version... Go

Displays audit comment history

Displays configuration logs

Compare changes between configuration versions.

```
7 category any ;
8 application [ dns google-base shutterfly
9 service application-default ;
10 source-hip any ;
11 destination-hip any ;
12 action allow ;
13 rule-type universal ;
14 description "Allows hosts in Users_Net zc
15 tag Internet ;
16 profile-setting {
17   profiles {
18     url-filtering default ;
19     virus default ;
20     spyware default ;
21   }
22 }
23 group-tag Internet ;
24 }
```

```
7 category any ;
8 application [ dns google-base shutterfly
9 service application-default ;
10 source-hip any ;
11 destination-hip any ;
12 action allow ;
13 rule-type universal ;
14 description "Allows hosts in Users_Net zc
15 tag Internet ;
16 profile-setting {
17   profiles {
18     url-filtering Corp URL Filter ;
19     virus Corp AV ;
20     spyware Corp AS ;
21   }
22 }
23 group-tag Internet ;
24 }
```

Configure a Security Policy Rule: Source Tab

Security Policy Rule

General | **Source** | Destination | Application | Service/URL Category | Actions | Usage

Default is **Any**. You can add multiple addresses, address groups, external dynamic lists, or geographical regions.

☐ Any

☐ SOURCE ZONE ^

☐ Users_Net

☒ []

Danger

Extranet

Internet

Users_Net

☒ Any

☐ SOURCE ADDRESS ^

External Dynamic List

- Palo Alto Networks - Bulletproof IP addresses
- Palo Alto Networks - High risk IP addresses
- Palo Alto Networks - Known malicious IP address

Region

- 0.0.0.0-0.255.255.255 (Reserved(0.0.0.0-0.255.255.255.255.255))
- 10.0.0.0-10.255.255.255 (Reserved(10.0.0.0-10.255.255.255.255.255))
- 100.64.0.0-100.127.255.255 (Reserved(100.64.0.0-100.127.255.255.255.255))
- 127.0.0.0-127.255.255.255 (Reserved(127.0.0.0-127.255.255.255.255.255))
- 169.254.0.0-169.254.255.255 (Reserved(169.254.0.0-169.254.255.255.255.255))

☐ SOURCE USER ^

any

any

pre-logout

known-user

unknown

select

☐ SOURCE DEVICE ^

any

any

no-hip

quarantine

select

☐ Negate

Policy will match all source addresses that are not listed.

+ Add - Delete

+ Add - Delete

+ Add - Delete

+ Add - Delete

Configure a Security Policy Rule: Destination Tab

Security Policy Rule

General | Source | **Destination** | Application | Service/URL Category | Actions | Usage

select

☐ DESTINATION ZONE

☐ Internet

☒

Danger

Extranet

Internet

Users_Net

+ Add - Delete

☒ Any

☐ DESTINATION ADDRESS

External Dynamic List

- Palo Alto Networks - Bulletproof IP addresses
- Palo Alto Networks - High risk IP addresses
- Palo Alto Networks - Known malicious IP address

Region

- 0.0.0.0-0.255.255.255 (Reserved(0.0.0.0-0.255.255.255.255.255))
- 10.0.0.0-10.255.255.255 (Reserved(10.0.0.0-10.255.255.255.255.255))
- 100.64.0.0-100.127.255.255 (Reserved(100.64.0.0-100.127.255.255.255.255))
- 127.0.0.0-127.255.255.255 (Reserved(127.0.0.0-127.255.255.255.255.255))
- 169.254.0.0-169.254.255.255 (Reserved(169.254.0.0-169.254.255.255.255.255))

+ Add - Delete

☐ Negate

any

☐ DESTINATION DEVICE

any

quarantine

select

+ Add - Delete

Default is **Any**. You can add multiple addresses, address groups, external dynamic lists, or geographical regions.

any
quarantine
select

Policy will match all source addresses that are not listed.

Configure a Security Policy Rule: Application Tab

Default is **Any**. You should add specific applications as match criteria.

The screenshot shows the 'Security Policy Rule' configuration window with the 'Application' tab selected. The 'Any' checkbox is unchecked, and the 'APPLICATIONS' section is expanded. The 'adobe-connectnow-base' application is selected with a checkmark. To the right, the 'DEPENDS ON' section is expanded, showing a list of dependencies: 'flash', 'rtmp', 'rtmpt', 'ssl', and 'web-browsing', all of which are checked. A callout box with the text 'Add application, then check for/add application dependencies.' has two arrows pointing to the 'adobe-connectnow-base' application and the 'DEPENDS ON' section. At the bottom right, there are two buttons: 'Add To Current Rule' and 'Add To Existing Rule', which are highlighted with a black border. The top of the window shows tabs for 'General', 'Source', 'Destination', 'Application', 'Service/URL Category', 'Actions', and 'Usage'. The bottom left has '+ Add' and '- Delete' buttons.

Security Policy Rule

General | Source | Destination | **Application** | Service/URL Category | Actions | Usage

☐ Any

☒ APPLICATIONS ^

☒ adobe-connectnow-base

Add application, then check for/add application dependencies.

☒ DEPENDS ON ^

- ☒ flash
- ☒ rtmp
- ☒ rtmpt
- ☒ ssl
- ☒ web-browsing

+ Add - Delete

Add To Current Rule Add To Existing Rule

Unresolved Dependencies Reported During a

	NAME	TAGS	TYPE	Source				Destination			APPLICATION	SERVICE	ACTION
				ZONE	ADDRESS	USER	DEVICE	ZONE	ADDRESS	DEVICE			
1	Users_to_Internet	Internet	universal	Users_Net	any	any	any	Internet	any	any	adobe-connectnow-base	application-default	Allow

Commit Status

Operation Commit

Status Completed

Result Successful

Details Partial changes to commit: changes to configuration by administrators: admin
Changes to policy and objects configuration
Configuration committed successfully

Commit **App Dependency**

RULE	COUNT
Users_to_Internet	5

1 item → X

APP	DETAIL
adobe-connectnow-base	• adobe-connectnow-base requires flash to be allowed. • adobe-connectnow-base requires rtmp to be allowed. • adobe-connectnow-base requires rtmp to be allowed. • adobe-connectnow-base requires ssl to be allowed. • adobe-connectnow-base requires web-browsing to be allowed.

1 item → X

Missing application dependencies

- A commit determines if application dependencies in *any* rule are satisfied by *any* rule.
- Unresolved dependencies are reported per rule.
- Click rule's **Count** number to view unresolved dependencies.
- Click **<rule_name>** to open and edit rule.

Configure a Security Policy Rule: Service/URL Category Tab

Default is **application-default**. You can add one or more services.

Default is **Any**. You can add multiple URL categories.

General | Source | Destination | Application | **Service/URL Category** | Actions | Usage

application-default

☐ SERVICE ^

application-default
any
select

Service

- service-http
- service-https

New Service Service Group

+ Add - Delete

☒ Any

☐ URL CATEGORY ^

External Dynamic Lists

- panw-auth-portal-exclude-list

Palo Alto Networks

- abortion
- abused-drugs
- adult
- alcohol-and-tobacco
- auctions

+ Add - Delete

Configure a New Service Definition

Objects > Services > Add

- Service definitions are assigned ports.
- Services limit ports that applications can use.
- service-http and service-https are the only predefined services.

Service

Name

Description

Protocol ☒ TCP ☐ UDP

Destination Port

Source Port

Port can be a single port #, range (1-65535), or comma separated (80, 8080, 443)

Session Timeout ☒ Inherit from application ☐ Override

Tags

Security Policy Rule

General | Source | Destination | Application | **Service/URL Category** | Actions | Usage

☐	SERVICE ^
☐	 service-SMTP

Use new service.

For a Tech Doc about this topic, log into Live and search for "Objects > Services."

Configure a Security Policy Rule: Actions Settings

Security Policy Rule

General | Source | Destination | Application | **Actions** | Usage

Action Setting

Action: Allow

☐ Send ICMP Unreach

Profile Setting

Profile Type: None

Log Setting

☐ Log at Session Start

☒ Log at Session End

Log Forwarding: None

Other Settings

Schedule: None

QoS Marking: None

☐ Disable Server Respon

Available with "drop" and all "reset" actions

Optional: Add session start for troubleshooting.

Can schedule when the rule is active

Schedule Security Policy Rules

- Policy rules may be enforced on only specific days and time periods.
- Use 24-hour time format.
- Can specify:
 - Daily
 - Days of week
 - Calendar days

Objects > Schedules > Add

Schedule

Name Policy_Rule_Enforcement_Times

Recurrence Daily

START TIME	END TIME
08:00	18:00

Recurrence options: Daily, Weekly, Non-recurring

+ Add - Delete

Create a schedule with one or more start and end times.

Policies > Security > <select_rule> > Actions

Other Settings

Schedule Policy_Rule_Enforcement_Times

QoS Marking None

Policy_Rule_Enforcement_Times

New Schedule

Apply schedule to a rule.

Configure a Security Policy Rule: Usage Settings

The screenshot shows the 'Usage' tab of a 'Security Policy Rule' configuration page. The page is divided into two main sections: 'Basics' and 'Activity' on the left, and 'Applications' and 'Traffic (past 30 days)' on the right. Callouts provide additional context for specific data points.

Security Policy Rule

General | Source | Destination | Application | Service/URL Category | Actions | **Usage**

Basics

- Rule Created 2020-07-14 17:59:46
- Last Edited 2020-07-14 20:44:04

Activity

- Hit Count 6331
- First Hit 0 days ago 2020-07-14 18:17:03
- Last Hit 0 days ago 2020-07-14 21:22:21

Applications

- Applications Seen 6
- Last App Seen 0 days ago
- [Compare Applications & Applications Seen](#)

Traffic (past 30 days)

- Bytes 1.7M

Callouts:

- View when rule was created and last modified. (points to Rule Created and Last Edited)
- View activity that matches rule. (points to Hit Count, First Hit, and Last Hit)
- Provides tools to migrate from port-based rules (points to Compare Applications & Applications Seen)
- View amount of traffic, in bytes, over the past 30 days. (points to Bytes 1.7M)

For a Tech Doc about this topic, log into Live and search for "Monitor Policy Rule Usage."

Enable Intrazone and Interzone Logging

Policies > Security > <select_default_rule>

4	intrazone-default	none	intrazone	any	any	any
5	interzone-default	none	interzone	any	any	any

⊕ Add ⊖ Delete 🔄 Clone ⚙️ Override ⬅️ Revert ✅ Enable ❌ Disable ➡️ Move ▾

- Traffic matching default rules normally is not logged.
- Could log for visibility and troubleshooting purposes.

Security Policy Rule

General | **Actions**

Action Setting

Action Deny ▾

☐ Send ICMP Unreachable

Profile Setting

Profile Type None ▾

Log Setting

☐ Log at Session Start

☒ Log at Session End

Log Forwarding None ▾

Find Unused Security Policy Rules

- Remove unused rules to:
 - Increase firewall operational efficiency
 - Simplify rule management
- Firewall tracks rules unused since last time the data plane restarted.

Policies > Security

	NAME	TAGS	TYPE	Source				Destination			APPLICATION	SERVICE	ACTION	HIT COUNT
				ZONE	ADDRESS	USER	DEVICE	ZONE	ADDRESS	DEVICE				
1	Users_to_Internet	Internet	universal	Users_Net	any	any	any	Internet	any	any	adobe-connectnow-base	application-default	Allow	77057
2	Users_to_Extranet	none	universal	Users_Net	any	any	any	Extranet	any	any	any	application-default	Allow	2521
3	Extranet_to_Internet	none	universal	Extranet	any	any	any	Internet	any	any	any	application-default	Allow	0
4	intrazone-default	none	intrazone	any	any	any	any	(intrazone)	any	any	any	any	Allow	83256
5	interzone-default	none	interzone	any	any	any	any	any	any	any	any	any	Deny	-

Unused rules highlighted

Highlight Unused Rules

Rule Usage Filter

PA-VM DASHBOARD ACC MONITOR **POLICIES** OBJECTS NETWORK DEVICE Commit ?

Security

- NAT
- QoS
- Policy Based Forwarding

Policy Optimizer

- New App Viewer 1+
- Rules Without App Controls 2
- Unused Apps 0
- Log Forwarding for Security Services
- Rule Usage**
- Unused in 30 days 5
- Unused in 90 days 5
- Unused 5

Rule Usage

These rules require prompt attention to prevent unused applications from accessing your network. These are security policy rules in your rulebase that no traffic has matched over the selected time period. Evaluate these rules and if you don't need them, delete them from the rulebase to reduce your attack surface and stop allowing applications that you don't use. If you need to allow one or a few applications on an unused rule, remove the unused applications. If the rule is for applications that are only used at specific time periods such as quarterly meetings, configure a schedule for the rule so that it's only active when you need it.

Timeframe **All time** Usage **Any** ☐ Exclude rules reset during the last 90 days

8 items

	NAME	HIT	LAST HIT	RESET DATE	MODIFIED	CREATED
5	Allow_UserNet_to_E...	199	2022-01-13 15:26:39	2022-01-12 16:19:26	-	2022-01-12 16:17:25
6	Allow-Internet-Access	549	2022-01-13 15:26:30	2022-01-12 16:19:26	-	2022-01-12 16:17:25
7	DMZ-to-Inside	0	-	-	-	2022-01-12 16:17:25
8	intrazone-default	20983	2022-01-13 15:24:38	2022-01-12 15:00:44	-	2022-01-12 14:57:59

Object : Addresses + Delete Enable Disable PDF/CSV Reset Rule Hit Counter Tag UnTag

For a Tech Doc about this topic, log into Live and search for "Sorting and Filtering Security Policy Rules."

Create an Address Object

Objects > Addresses > Add

Address

Name: Bureau-121

Description: Address object for Bureau-121

Type: IP Netmask

212.0.34.0/24 [Resolve](#)

Enter an IP address or a network using the slash notation (Ex. 192.168.80.150 or 192.168.80.0/24). You can also enter an IPv6 address or an IPv6 address with its prefix (Ex. 2001:db8:123:1::1 or 2001:db8:123:1::/64)

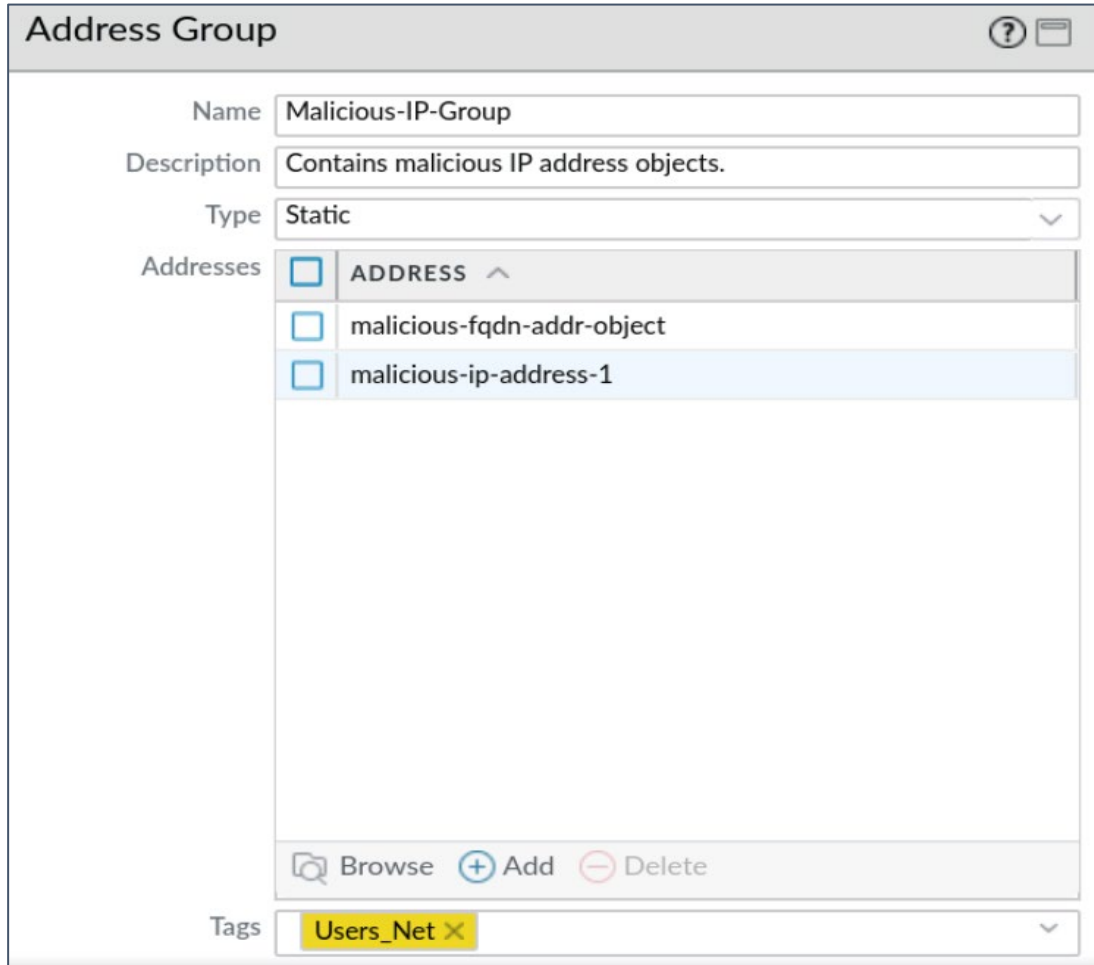
Tags: IP Netmask, IP Range, IP Wildcard Mask, FQDN

OK Cancel

- Address objects can represent:
 - A single IP address
 - An IP netmask
 - An IP address range
 - A specific set of addresses
 - An FQDN
- Add malicious IP addresses to an address object:
 - The list of known-bad IP addresses can change quickly.
- Address objects can be used in Security policy rules:
 - In source or destination address fields

Create a Static Address Group

Objects > Address Groups > Add

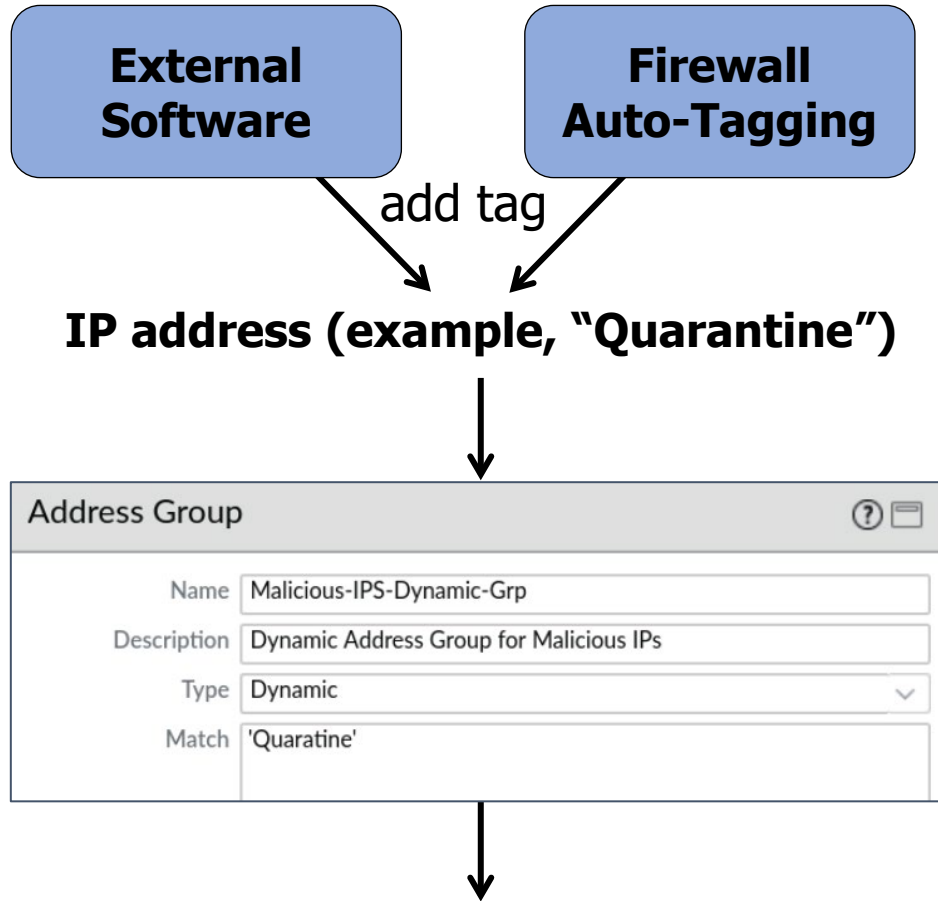


The screenshot shows the 'Address Group' configuration window. The 'Name' field is 'Malicious-IP-Group'. The 'Description' field is 'Contains malicious IP address objects.'. The 'Type' is set to 'Static'. Under the 'Addresses' section, there are three entries: 'ADDRESS' (with an expand/collapse icon), 'malicious-fqdn-addr-object', and 'malicious-ip-address-1'. At the bottom, there are 'Browse', 'Add', and 'Delete' buttons. The 'Tags' section shows 'Users_Net' with a close icon.

- Use to shorten and simplify a policy rule
- Membership must be manually updated
- Updates require a commit operation
- Can contain:
 - Address objects
 - Other address groups

For a Tech Doc about this topic, log into Live and search for "Objects > Address Groups."

Create a Dynamic Address Group



- Tagged IP addresses automatically added to group
- No commit necessary
- IP addresses in group used as match condition in rules

	NAME	TAGS	TYPE	Source			Destination		APPLICATION	SERVICE	ACTION
				ZONE	ADDRESS	USER	ZONE	ADDRESS			
1	Block-Known-Bad-IPs	Users_Net	universal	Users_Net	any	any	Internet	Malicious-IPS-Dynamic-Grp	any	any	Deny

Tags

Objects > Tags > Add

Tag

Name

Mail_Servers

Color

Turquoise Blue

Comments

Tag for Mail Servers on the Extranet

Security Policy Rule

General

Source

Destination

Application

Service/URL Category

Name

Protected_Mail_Servers

Rule Type

universal (default)

Description

Policy designed to protect th

Tags

Mail_Servers

Group Rules By Tag

Mail_Servers

Audit Comment

Mail Servers Policy created

Audit Comment Archive

- Use tags to visually search or use tag filters to find objects.
- Rules and objects can have multiple tags.

Filter for tag.

(tag/member eq 'Mail_Servers')

	NAME	TAGS	TYPE	ZONE
4	Protected_Mail_Servers	Mail_Servers	un	
5	intrazone-default	none	intrazone	any
6	interzone-default	none	interzone	any

Look for tag color.

Tag-Based Rule Groups

- Visually group rules based on tagging structure
- Can perform operational procedures within the selected tag group

Policies > Security

The screenshot displays the Palo Alto Networks Security Rule configuration interface. On the left, a sidebar lists rule groups: Internet (1), Extranet (1), Internet (1), and Mail_Servers (1). The main table shows a single rule, 'Users_to_Internet', with a priority of 1. A callout box points to the priority column, stating 'Maintains rule priority'. A context menu is open over the 'Group' dropdown in the bottom right, showing options: Change group of all rules, Move all rules in group, Delete all rules in group, and Clone all rules in group. The bottom toolbar includes buttons for Add, Delete, Clone, Override, Revert, Enable, Disable, Move, PDF/CSV, Highlight Unused Rules, View Rulebase as Groups (checked), Reset Rule Hit Counter, and a Group dropdown.

			NAME	TAGS	TYPE	Source			Destination		APPLICATION	SERVICE	ACTION
						ZONE	ADDRESS	USER	ZONE	ADDRESS			
Internet (1)	1	1	Users_to_Internet	Internet	universal	Users_Net	any	any	Internet	any	adobe-connectnow-base	application-default	Allow

Test Policy Functionality

Policies > Security

	NAME	TAGS	TYPE	Source				Destination			APPLICATION	SERVICE	ACTION	HIT COUNT
				ZONE	ADDRESS	USER	DEVICE	ZONE	ADDRESS	DEVICE				
1	Users_to_Extranet	Extranet	universal	Users_Net	any	any	any	Extranet	any	any	any	application-default	Allow	9806
2	Users_to_Internet	Internet	universal	Users_Net	any	any	any	Internet	any	any	web-browsing	application-default	Allow	77426
3	Extranet_to_Internet	Int										application-default	Allow	0

Test Security Policy Match

Test Configuration

Select Test

Security Policy Match

From

Users_Net

To

Internet

Source

192.168.1.20

Source Port

[1 - 65535]

Destination

8.8.8.8

Destination Port

80

Source User

None

Protocol

TCP

☐ show all potential match rules until first allow rule

Application

None

Uappid

[10000000 - 4294967295]

Category

None

☐ check hip mask

Source OS

None

Source Model

None

Source Vendor

None

Destination OS

None

Destination Model

None

Test Result

Allow-Internet-Access

Result Detail

NAME	VALUE
Name	Allow-Internet-Access
Index	6
From	Extranet
	Users_Net
Source	any
Source Region	none
To	Internet
Destination	any
Destination Region	none
User	any
source-device	any
destination-device	any
Category	any
Application Service	0:any/any/any/app-default
Action	allow
ICMP Unreachable	no
Terminal	yes

Test criteria

Policy details

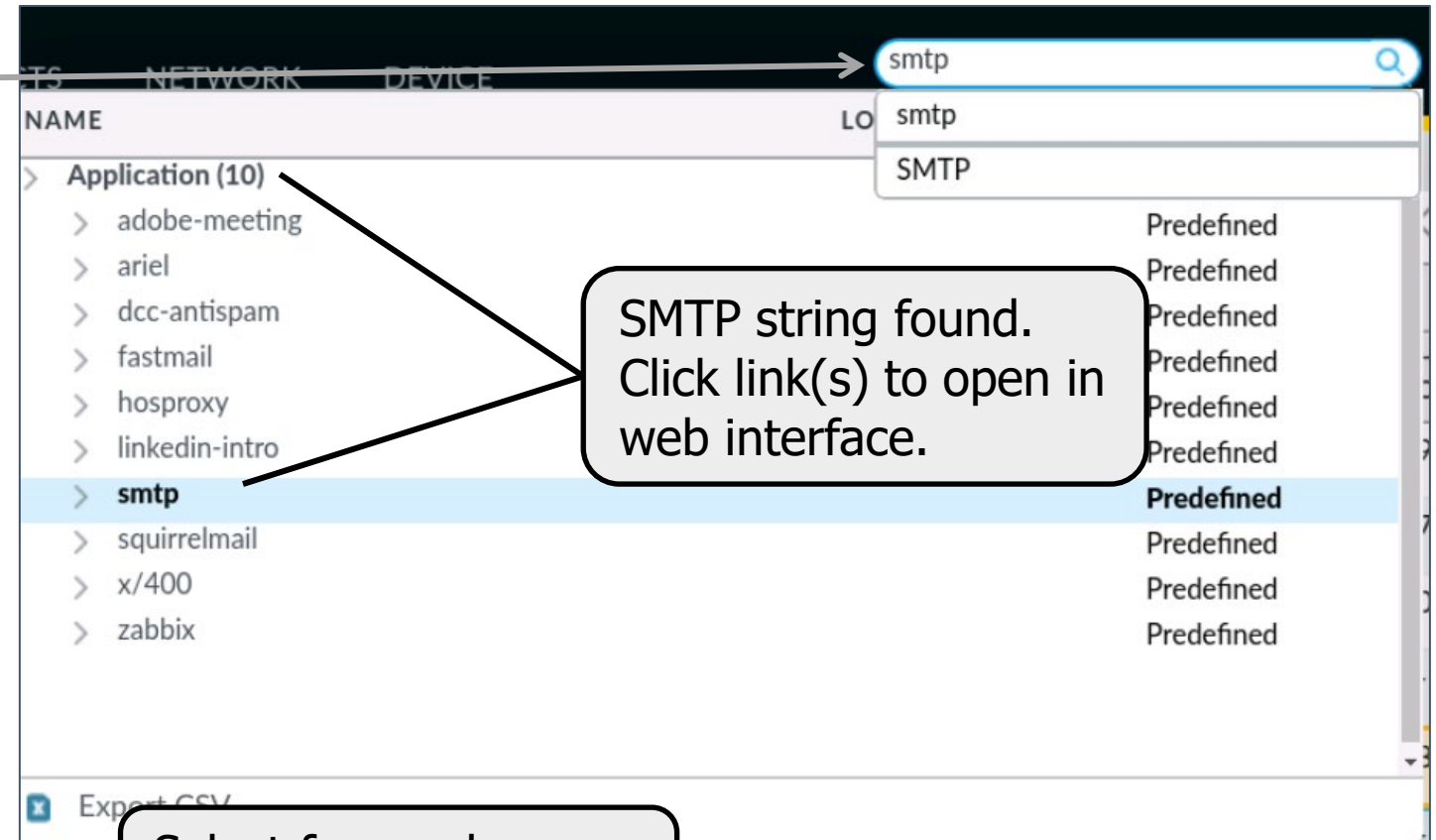
Test Policy Match

Use Global Find



- Search candidate configuration and content databases for occurrences of a string.
- Launch from **Search** or **Context** menu.

	NAME	TAGS	TYPE	ZONE
1	Users_to_Extranet	Extranet	Edit...	Users_Net
2	Users_to_Internet	Internet	Filter	Users_Net
3	Extranet_to_Internet	Internet	Global Find	Extranet



For a Tech Doc about this topic, log into Live and search for "Global Find."

View the Traffic Log

Monitor > Logs > Traffic

		RECEIVE TIME	TYPE	FROM ZONE	TO ZONE	SOURCE	DESTINATION	TO PORT	APPLICATION	ACTION	RULE	SESSION END REASON	BYTES
		07/14 19:45:48	end	Users_Net	Internet	192.168.1.20	216.58.194.142	80	google-base	allow	Users_to_Internet	tcp-fin	912
		07/14 19:44:13	end	Users_Net	Internet	192.168.1.20	172.217.1.142	80	google-base	allow	Users_to_Internet	tcp-fin	912
										allow	Users_to_Internet	tcp-fin	912

Detailed Log View

General

Session ID 127457

Action allow

Session Source from-policy

Host ID

Application google-base

Rule Users_to_Internet

Rule UUID e1a4b347-e1d4-45ba-8c97-5505e682e62f

Session End Reason tcp-fin

Category any

Device SN

IP Protocol tcp

Log Action

Source

Source User

Source 192.168.1.20

Source DAG

Country 192.168.0.0-192.168.255.255

Port 55648

Zone Users_Net

Interface ethernet1/2

NAT IP 203.0.113.20

NAT Port 28419

X-Forwarded-For IP 0.0.0.0

Destination

Destination User

Destination 172.217.9.14

Destination DAG

Country United States

Port 80

Zone Internet

Interface ethernet1/1

NAT IP 172.217.9.14

NAT Port 80

Flags

Captive Portal ☐

PCAP	RECEIVE TIME	TYPE	APPLICAT...	ACTION	RULE	RULE UUID	BY...	SEVERI...	CATEG...	URL CATEG...	VERDI...	URL	FILE NAME
	2020/07/14 19:42:47	end	google-base	allow	Users_t...	e1a4b...	912		any				

View details.

- Logs written to Traffic log.
- Each Security policy rule can log the start and/or end of each session.
- Default is to log at session end.
- Temporarily add session start for troubleshooting.

Module Summary

Now that you have completed this module, you should be able to:

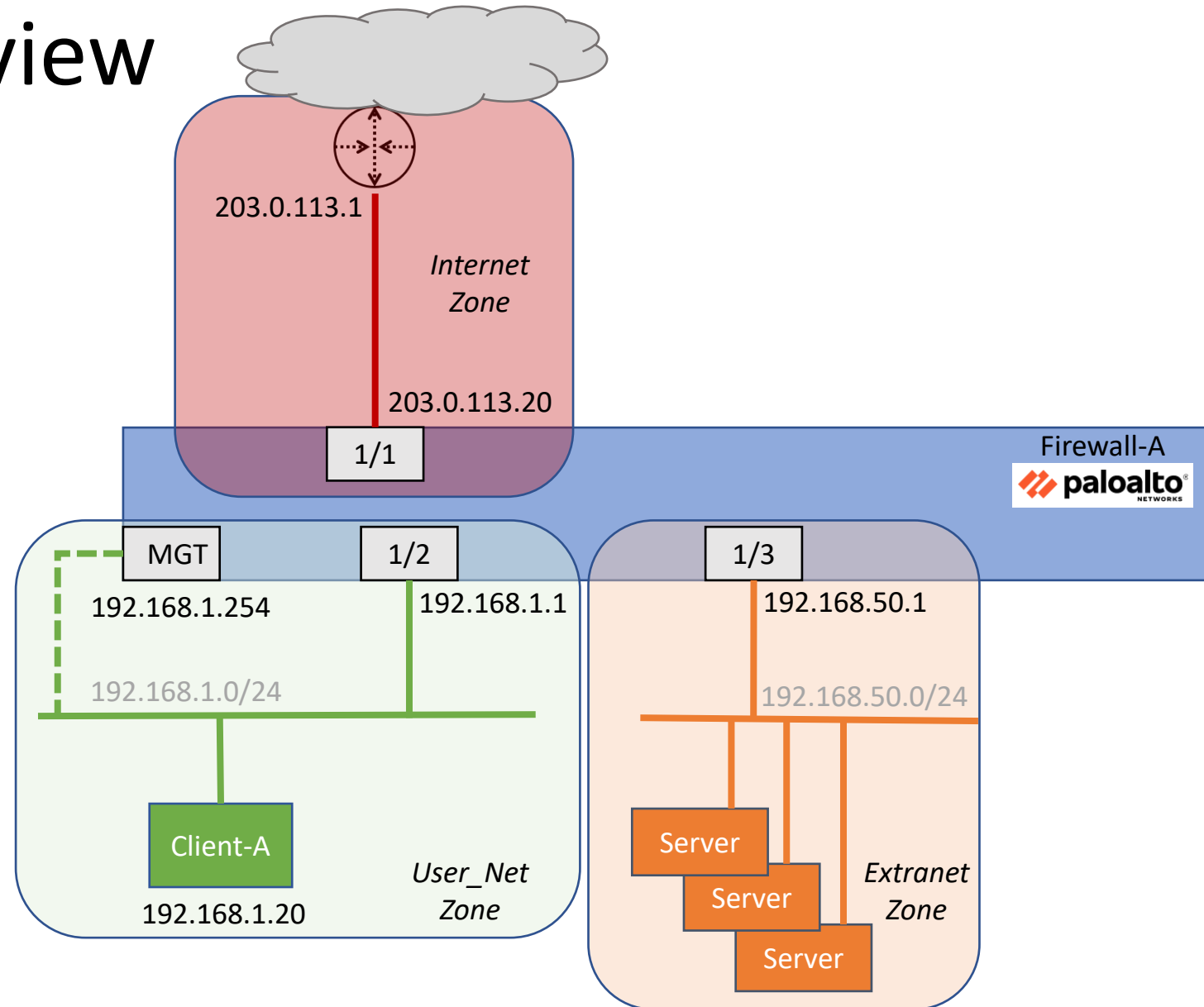
- Describe Security policy concepts and operation
- Configure a Security policy rule
- Manage a Security policy



Questions



Lab 6: Overview



Lab 6: Creating and Managing Security Policy Rules

- Configure a Security policy rule to allow access from Users_Net to Extranet
- Test access from client to Extranet servers
- View the Traffic log
- Examine policy Rule Hit Count
- Reset rule hit counts
- Customize policy tables
- Enable intrazone and interzone logging
- Create Security Policy Rules to Internet Zone
- Create Block Rules for Known-Bad IP Addresses



**Protecting our
digital way
of life.**

End Content

Unit 6 Homework

- Always check Blackboard for assignments and when they are due.
- Unit 6 Lab: Netlabs Lab 5 – Create and Manage Security Policy Rules
- Unit 6 Quiz